

flea

NEWS **71** *December 2012*



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FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

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Editorial

Dear Flea News Reader,

Sadly, I include the obituary of Glenn Haas. Dr. Haas worked on a variety of innovative research on flea ecology, especially on the suitability of different hosts. This line of research is of interest in the attempt to understand the nature of flea species, that is, "why do these species of flea exist?", or as the ecologist Hutchinson asked, "why are there so many kinds of animals?"

Hutchinson asked his question about the paradox of many species of plankton in a seemingly homogeneous open ocean. Siphonapterists have a different problem: we know of many, heterogeneous, yet plausible, reasons as to why certain flea species maintain a strong association with certain hosts. These reasons range from simple co-occurrence in the same area, to life cycle matching of parasite and host, to hypothetical differences in blood characteristics.

Nonetheless, the explanation for why there is the diversity of flea species and subspecies we find in nature has remained elusive. That is one reason why techniques such as artificial hosts (see the Announcements section) are so interesting. Researchers may be able to construct artificial bloods to determine, experimentally, if flea species are specialized for different host bloods.

On another subject, the reader may wonder how we group the 'Featured Research' citations, since they often are not alphabetical. We group these articles in part by alphabet, but in part by common, secondary authors. This allows you to see together the varied investigations of a laboratory, even with papers listing different lead authors. In addition, you may notice repetitions of some citations in the different volumes of Flea News. This happens for various reasons. One is that preprint and first mentions of a paper may have incomplete citation information, but we list them anyway. When the entire citation is available, we may list the paper again with the updated information. This section may contain papers no longer current if they are of interest.

I thank J. Kucera and R.E. Lewis for help with this volume of Flea News. Thanks to everyone

who contributed material for this Flea News. Best wishes to you all for a happy new year 2013!

- R.L. Bossard, Editor, *Flea News*

Reference: G. E. Hutchinson. 1959. Homage to Santa Rosalia or Why Are There So Many Kinds of Animals. *The American Naturalist* Vol. 93, No. 870: 145-159.

Announcements

The International Symposium on Ectoparasites of Pets will be held in Munich, Germany, April 7-9, 2013. Abstract submissions are due by January 31 st . Contact organizers at info@isep2013.net. Further information is available at the ISEP website <http://www.isep2013.net/> .

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Dear Sir or Madam:

The Entomological Society of America (ESA) has become aware of a proposed 2013 meeting that is causing some confusion among our members and others in the entomological community. As you probably know, ESA is currently planning two major meetings: 1) the 2013 ESA Annual Meeting, entitled "Entomology 2013," to be held in Austin, TX, in November; and 2) the 2016 International Congress of Entomology, to be held in Orlando, FL (ICE 2016). However, we have been contacted by a third party that plans to hold a meeting in September 2013 with a name that is VERY SIMILAR to the ones listed above. Perhaps you have been notified, too. Please note that this other event is NOT supported by ESA, nor is it related to any ESA-sponsored event in any way.

The official websites for the two ESA-supported events listed above are:

<http://www.entsoc.org/entomology2013> and

<http://www.ice2016orlando.org>.

If you receive a solicitation to participate or register with a meeting that sounds similar to the official ESA events, be sure to check the source and sender of the email. Although ESA members and others are free to participate in any meeting they choose, please be aware that the similar-sounding meeting is not related to, nor sponsored by, ESA. If you choose to commit to other meetings, be sure to inquire about how your name and research will be used, and what fees are involved. Official correspondence on Entomology 2013 and ICE 2016 will come from ESA; with ICE 2016 correspondence over co-chairs Walter Leal and Alvin Simmons names.

If you have any questions about ESA's upcoming events, please contact Rosina Romano, ESA Director of Meetings, at rromano@entsoc.org.

Please share this information with your colleagues, affiliates and friends around the world, to help them be aware as well.

Thanks for your assistance.

Rob Wiedenmann

Entomological Society of America 2013 President

Research in Progress

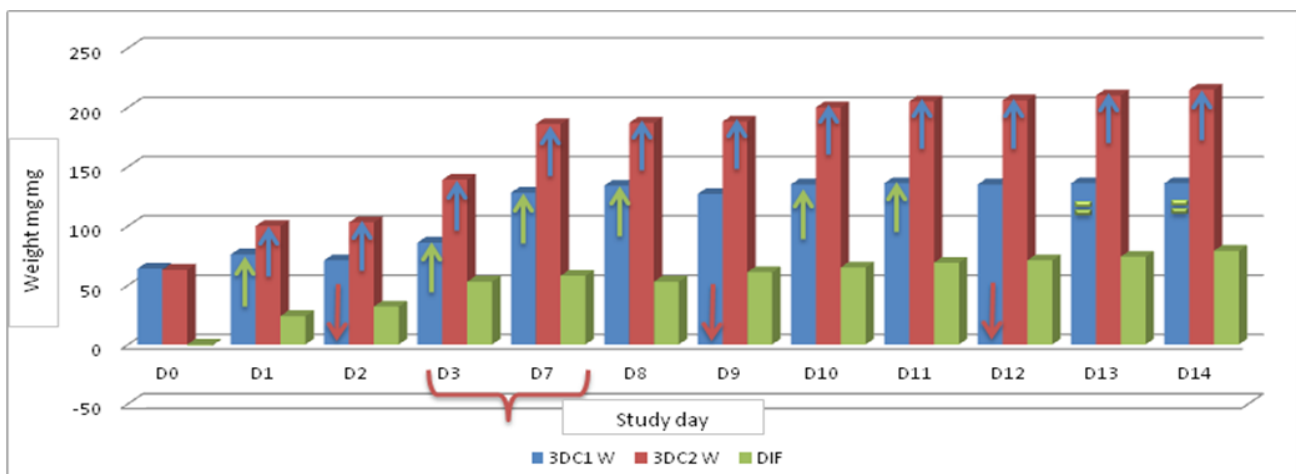
Preliminary results using an artificial host: Colonized cat fleas gain weight faster on cattle, than dog, blood.

The Artificial Dog is an artificial host, flea-feeding system to breed fleas. It seems that only a *Ctenocephalides felis* colony named "Georgi", a special strain of the cat flea isolated from cattle, can feed in this system. Cattle blood is commonly used to feed fleas in Monheim lab due to its easy availability.

In 1988, Wade et al. affirmed that no significant differences were observed in the survival of fleas fed with cattle blood or dog blood. Indeed, *C. felis* is commonly regarded as a non-host-specific parasite. Our strain colony is fed with cattle blood since 1994 and has never been mixed. Surprisingly, the colony has recently fed with dog blood showing lower feeding rates. This observation led us to pose a question: could feeding a colony with cattle blood for 18 years have modified its blood preference?

To consider that question a study was performed: two groups of fleas (approximately 150 per group) of *C. felis* "Georgi" colony were fed daily by the Artificial Dog system for 15 days. From day 0 (D0) to day 14 (D14), one group (3DC1) was fed with dog blood (10ml per day), the other group (3DC2) was fed with cattle blood (10ml per day). Groups were weighed daily, and mortality rate determined. At the beginning of the study (D0), both groups weighed the same. From D1 to D3, group 3DC2 (fed with cattle blood) showed a higher weight increase than group 3DC1 (fed with dog blood) (Graph 1). The mortality rate was a bit higher in group 3DC1, however, no statistically significant differences were observed between the two groups.

Finally, we observed a big difference between the feeding rates in the two groups. Feeding *C. felis* "Georgi" colony with cattle blood for many generations may have modified its blood preference. In this summary, we report only preliminary data of our study, and we are looking forward to hearing your experience (submitted by Ettore Napoli).



Graph 1. Weight and weight differences in cat fleas of group 3DC1 (dog blood) and in group 3DC2 (cattle blood) during the course of the study.

Editor's note:

It is an interesting observation that cat flea gain weight faster on cattle blood, than on what we would expect to be a normal diet of dog blood. Bloods of various animals vary in cell counts and characteristics, physical parameters, fibrin levels, and so on, which could affect the nutrition of a blood for a particular flea.

It is likely that the fleas, the "Georgi" colony in particular, colonized for so many years, may have adapted to both cattle blood and artificial feeding. The artificial feeding system could shed light on blood suitability, which may help determine host-association of fleas in nature. It is wonderful to hear from other researchers working with cat fleas, and especially someone who is rearing a colony on an artificial feeding system. It is impressive that your colony has been maintained so long without a living host. This is exactly the kind of research overview that we like to include, so that other flea researchers have some perspective on the flea work that is being conducted by colleagues around the world.

It will be exciting to see how your study progresses and we look forward to reading your final conclusions.

- Editors, Flea News

References:

Hinkle, N.C., P.G. Koehler, and R.S. Patterson. Flea Rearing in Vivo and in Vitro for Basic and Applied Research, in *Advances in Insect Rearing for Research and Pest Management*, Anderson, T. E., et al. (eds.), Westview Press, Boulder, 1992, pp. 119-130.

Wade E. S. and Georgi R. J. 1988. Survival and reproduction of Artificially Fed Cat Fleas, *C. felis* Bouché (Siphonaptera: Pulicidae). *J. Med. Entomol.* 25(3): 186-190.

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A hypothesis concerning factors diversifying bat fleas (Siphonaptera: Ischnopsyllidae) at the generic level.

To understand a taxon, we would like to know, first, how it is different from other, similar taxa. Second, the factors involved in creating its diversity. The hypothesis I entertain here is an approach to the second question. I hypothesize these factors have been important in creating genera of bat fleas: biogeographic isolation, temperature (hot or cold), and bat migration and metabolism (active, hibernation, or torpor). I estimate that there are, with seven biogeographic provinces (Afrotropical, Australian, Indomalayan, Nearctic, Neotropical, Oceanian, Palearctic) x two temperatures x two metabolisms (migratory or non-migratory), giving 28 genera of bat fleas worldwide. There are actually only 19 genera. At least in the northerly part of the Nearctic, the estimate seems consistent with observed numbers of bat flea genera (four): *Hormopsylla* occur on tropical bats; *Nycteridopsylla* are cold, temperate fleas on hibernating bats; *Myodopsylla* are hot, temperate fleas on active bats; and *Sternopsylla* are temperate fleas on migrating bats (submitted by R.L. Bossard).

Featured Research

Abdissa, A., Diatta, G., Mediannikov, O., Raoult, D., & Trape, J.-F. (2012). *Rickettsia felis* in fleas, southern Ethiopia, 2010. *Emerging Infectious Diseases*, 18(8), 1385+.

Acosta R. 2011. Distributional data and taxonomic notes on the flea *Strepsylla* (Siphonaptera: Ctenophthalmida: Neopsylla: Phallacropsyllini). *Revista Mexicana de Biodiversidad*, 82: 1154 – 1162.

Falcón-Ordaz J., R. Acosta, J. A. Fernández y G. Lira-Guerrero. 2012. Helmintos y sifonápteros parásitos de cinco especies de roedores en localidades de la Cuenca Oriental, en el centro de México. *Acta Zoológica Mexicana (nueva serie)*, 28(2): 287-304.

Guzmán-Cornejo, C., L. García Prieto, R. Acosta-Gutiérrez, J. Falcón-Ordaz y L. León-Paniagua. 2012. Metazoarios parásitos de *Tlacuatzin canescens* y *Marmosa mexicana* (Mammalia: Didelphimorphia) de México. *Revista Mexicana de Biodiversidad*.

Beaucournu J.-C. et al., 2012. Description de *Tunga bonneti* n. sp. du Chili (Siphonaptera : Tungidae) et notes sur sa spécificité, sa chorologie, son dermeco et sa phénologie. *Parasite*, 19: 207-216. (*Tunga bonneti* Beaucournu & Gonzalez-Acuna)

Lemaire J.-M. & Beaucournu J.-C., 2012. Une localité nouvelle et un hôte nouveau pour la Puce d'Oiseau *Ceratophyllus vagabundus alpestris* Jordan, 1926. *Biocosme mésogéen*, 29 : 37-41.

Perrichot V., Beaucournu J.-C. & Velten Jürgen , 2012. First extinct genus of a flea (Siphonaptera : Pulicidae) in Miocene amber from the Dominican Republic. *Zootaxa*, 3438 : 54-62. (*Eospilopsyllus kobberti* Beaucournu & Perrichot)

Beaucournu J.-C. et al., 2012. Le genre *Tunga* Jarocki, 1838 (Siphonaptera : Tungidae). I - Taxonomie, Phylogénie, Ecologie, Rôle pathogène. *Parasite*, 19 : 297-308.

Laudisoit A., Prié V. & Beaucournu J.-C. 2012. Une nouvelle espèce de *Lagaropsylla* Jordan & Rothschild, 1921 de Madagascar (Insecta, Siphonaptera, Ischnopsyllidae). *Zoosystema* 34 (4): 737–744. [*A new species of Lagaropsylla. Jordan & Rothschild, 1921 from Madagascar (Insecta, Siphonaptera, Ischnopsyllidae).*]

This paper describes a new species of the genus *Lagaropsylla* Jordan & Rothschild, 1921, collected in the Makay Mountains in Madagascar, on the bat *Otomops madagascariensis* Dorst, 1953. Male differs from *L. obliqua* Smit, 1957 and *L. hoogstraali* Smit, 1957 by the *hamulus* shape, which is markedly higher than large, exhibiting a small “tooth” backwards of the apex; and differs from *L. consularis* Smit, 1957 by the preoral tuber, the pronotal comb and the *ductus ejaculatorius* shapes. Female has a sclerite at the *ductus bursae* base and differs from *L. consularis* by the preoral tuber shape and rather short spines on the comb. The new species' discovery is explained, in one hand, by the rarity of the host, and, in the other hand, by the lack of previous bats and ectoparasites studies in the remote Makay Mountains. A new parasite-host association between *Araeopsylla martialis* (Rothschild, 1903)

and *O. madagascariensis* is mentioned.

Sarah N. Bevins, Scott Carver, Erin E. Boydston, Lisa M. Lyren, Mat Alldredge, Kenneth A. Logan, Seth P. D. Riley, Robert N. Fisher, T. Winston Vickers, Walter Boyce, Mo Salman, Michael R. Lappin. 2012. Three Pathogens in Sympatric Populations of Pumas, Bobcats, and Domestic Cats: Implications for Infectious Disease Transmission. Plos One , vol. 7, no. 2.

Anthropogenic landscape change can lead to increased opportunities for pathogen transmission between domestic and non-domestic animals. Pumas, bobcats, and domestic cats are sympatric in many areas of North America and share many of the same pathogens, some of which are zoonotic. We analyzed bobcat, puma, and feral domestic cat samples collected from targeted geographic areas. We examined exposure to three pathogens that are taxonomically diverse (bacterial, protozoal, viral), that incorporate multiple transmission strategies (vector-borne, environmental exposure/ingestion, and direct contact), and that vary in species-specificity. *Bartonella* spp., Feline Immunodeficiency Virus (FIV), and *Toxoplasma gondii* IgG were detected in all three species with mean respective prevalence as follows: puma 16%, 41% and 75%; bobcat 31%, 22% and 43%; domestic cat 45%, 10% and 1%. *Bartonella* spp. were highly prevalent among domestic cats in Southern California compared to other cohort groups. Feline Immunodeficiency Virus exposure was primarily associated with species and age, and was not influenced by geographic location. Pumas were more likely to be infected with FIV than bobcats, with domestic cats having the lowest infection rate. *Toxoplasma gondii* seroprevalence was high in both pumas and bobcats across all sites; in contrast, few domestic cats were seropositive, despite the fact that feral, free ranging domestic cats were targeted in this study. Interestingly, a directly transmitted species-specific disease (FIV) was not associated with geographic location, while exposure to indirectly transmitted diseases – vector-borne for *Bartonella* spp. and ingestion of oocysts via infected prey or environmental exposure for *T. gondii* – varied significantly by site. Pathogens transmitted by direct contact may be more dependent upon individual behaviors and intra-specific encounters. Future studies will integrate host density, as well as landscape features, to better understand the mechanisms driving disease exposure and to predict zones of cross-species pathogen transmission among wild and domestic felids.

E Bouhsira; E Lienard; P Jacquet; S Warin; V Kaltsatos; L Baduel; M Franc. 2012. Environmentally relevant mixtures in cumulative assessments: an acute study of toxicokinetics and effects on motor activity in rats exposed to a mixture of pyrethroids. Toxicol. Sci. 130, 309.

Brianti E, Gaglio G, Napoli E, Giannetto S, Dantas-Torres F, Bain O, Otranto D. 2012. New insights into the ecology and biology of *Acanthocheilonema reconditum* (Grassi, 1889) causing canine subcutaneous filariasis. Parasitology. Apr;139(4):530-6.

In spite of its wide distribution among dogs and the evidence of its implication as a zoonotic agent, scant information is available on the biology of *Acanthocheilonema reconditum* (Spirurida, Onchocercidae). In this study, blood samples from 152 Sicilian dogs were examined for *A. reconditum microfilariae* at the beginning of the study and 1 year later. The periodicity of microfilaraemia was investigated by bleeding 2 highly microfilaraemic dogs twice a day for 10 days and, later on, every 2 weeks for 1 year and a third animal every 3 h for 96 h. Fleas and ticks infesting dogs were collected and dissected for the detection of *A. reconditum* larvae. The prevalence of infestation was 11.2% (17/152) and 13.3% (16/120) at the beginning and at the end of the study, with a 1 year cumulative

incidence of 5-9%. Although dogs bled twice a day showed a higher number of microfilariae in most of the morning samples, the absence of any circadian rhythm was suggested by data of the third experiment conducted by bleeding a dog every 3 h for 4 days. *A. reconditum* developing forms were detected in 5.1% (4/78) of dissected fleas, but not in any of the 272 ticks. The study provides new insights into the biology and ecology of this dog filarioid in its definitive and intermediate hosts.

BRINKERHOFF, ROBERT JORIS. 2008. MAMMAL AND FLEA OCCURRENCE IN ASSOCIATION WITH BLACK-TAILED PRAIRIE DOG (*CYNOMYS LUDOVICIANUS*) COLONIES: IMPLICATIONS FOR INTERSPECIFIC PLAGUE TRANSMISSION. Doctor of Philosophy thesis, University of Colorado. [selected abstracts]

Sylvatic plague is an enigmatic disease affecting over 200 mammalian species worldwide, yet its dynamics and ecology are poorly understood. Transmitted by fleas, plague was introduced into North America in the 19th century and threatens humans and wildlife species, most notably, the black-tailed prairie dog. This thesis focuses on elucidating potential mechanisms of plague transmission among mammals in Colorado by examining the community of alternate disease hosts and vectors in which plague exists in four ways. First, I examined spatial and temporal patterns of plague emergence in mammalian species most likely to acquire infection. Second, I described and evaluated occurrence patterns of fleas on a variety of mammalian species associated with black-tailed prairie dog colonies. Third, I tested for interactions within and between flea species on black-tailed prairie dogs. Fourth, I analyzed the population genetic structure of fleas collected from nine prairie dog colonies with different histories of plague occurrence. I detected universally low levels of plague exposure in mammals associated with prairie dog colonies, suggesting that plague is uncommon between epizootic events. Patterns of flea species occurrence among mammalian species suggest that the small rodents hypothesized to be involved in plague dynamics may play only a small role in plague transmission, whereas predators of prairie dogs may be responsible for harboring and transporting plague-infected fleas among colonies. Flea species apparently do not compete for resources and the presence of one flea species on a host may facilitate colonization by other flea species. Finally, the population genetic structure of prairie dog fleas is reflective of plague history at a given site and lends insights into patterns of recolonization following plague events. Taken as a whole, these results provide a uniquely detailed assessment of patterns of host and vector occurrence in the context of plague and they challenge previously held beliefs about plague dynamics in wildlife communities.

Sylvatic plague, caused by the bacterium *Yersinia pestis*, is a flea-borne disease that primarily affects rodents but has been detected in over 200 mammalian species worldwide. Mammalian carnivores are routinely surveyed as sentinels of local plague activity, since most species of carnivores are apparently resilient to infection, showing few signs of morbidity. In Boulder County, Colorado, USA, periodic plague epizootic events occur in black-tailed prairie dogs; enzootic hosts are unidentified as are plague foci. For three years, I systematically sampled carnivores in two distinct habitat types to determine whether carnivores may play a role in maintenance or transmission of plague and to identify habitats associated with

increased plague prevalence. I sampled 83 individuals representing six species of carnivores and found only two that had been exposed to *Y. pestis*. The overall low rate of plague exposure in carnivores suggests that plague may be very rare in this study system, or that carnivores are poor sentinels of plague. In either case, I cannot draw any conclusions regarding habitat-associated plague foci or temporal changes in plague activity. Plague epizootics involving prairie dogs were confirmed in this study system during two of the three years of this study, and I therefore suggest that the use of carnivores to survey for plague may be of limited value.

Inter-specific transmission of *Yersinia pestis*, the causative agent for sylvatic plague, is thought to be predominantly reliant on fleas. A number of species of mammals and fleas exhibit infection with *Y. pestis*, yet definitive vectors and reservoirs are unidentified in most plague systems. In western North America, blacktailed prairie dogs are particularly susceptible to plague infection and suffer high rates of mortality when plague infects a colony. I collected fleas from 13 species of mammals associated with black-tailed prairie dog colonies to determine which host species are most likely to spread plague-infected fleas to or from prairie dogs. I calculated community similarity indices and performed cluster and ordination analyses to determine which mammalian species had the most similar flea assemblages. Analyses indicated that the flea assemblages of prairie dogs diverged from those of small rodents but were very similar to certain species of carnivores as well as desert cottontail rabbits. These results are inconsistent with the paradigm that small rodents serve as reservoirs for *Y. pestis* and are sources for infection in prairie dogs. Rather I conclude that other mammals, including carnivores and lagomorphs are more likely to spread plague-infected fleas to and from prairie dogs.

Plague is a flea-borne, bacterial disease that threatens black-tailed prairie dogs in western North America. The identities of flea species responsible for spreading plague are unknown, but the prairie dog flea, *Oropsylla hirsuta*, is assumed to play a role in intraspecific plague transmission. I examined the population genetic structure of *O. hirsuta* fleas collected from nine prairie dog colonies, five of which had experienced a plague-associated die-off in Boulder County, Colorado. I expected that fleas collected from colonies with a history of recent plague activity would have less genetic diversity and would be less distinctive genetically than fleas collected from colonies with no recent history of plague. I also expected that genetic differentiation would be dependent on distance, with fleas from recently plague-affected colonies most similar to their nearest neighbors. Three plague-negative colonies showed significant differentiation from all other colonies, as did two of the five recently plague-positive colonies. I found that all five recently plagued colonies showed signs of recent population expansion whereas two of the four plague-negative colonies showed patterns consistent with stable population sizes. I did not detect a signal of isolation by distance, suggesting that *O. hirsuta* may be able to disperse relatively quickly over long distances. These results suggest that *O. hirsuta* populations are affected by plague-induced prairie dog die-offs and that flea dispersal among prairie dog colonies may be accomplished by multiple mechanisms.

Behavioral, genetic, and immune variation within a host population may lead

to aggregation of parasites whereby a small proportion of hosts harbor a majority of parasites. In situations where two or more parasite species infect the same host population there is the potential for interaction among parasites that could potentially influence patterns of aggregation through either competition or facilitation. I studied the occurrence and abundance patterns of two congeneric flea species on black tailed prairie dog (*Cynomys ludovicianus*) hosts to test for interactions among parasite species. I live-trapped prairie dogs on ten sites in Boulder County, Colorado and collected their fleas. I found a non-random, positive association between the two flea species, *Oropsylla hirsuta* and *O. tuberculata cynomuris*; hosts with high loads of one flea species had high loads of the second species. This result suggests that there is no interspecific competition among fleas on prairie dog hosts. Host weight had a weak negative relationship to flea load and host sex did not influence flea load, though there were slight differences in flea prevalence and abundance between male and female *C. ludovicianus*. Although genetic and behavioral variation among hosts may predispose certain individuals to infection, these results indicate apparent facilitation among flea species that may result from immune suppression or other flea-mediated factors.

Megan M. Friggens, Paul Beier. 2010. Anthropogenic disturbance and the risk of flea-borne disease transmission. *Oecologia*, vol. 164, no. 3, pp. 809-820.

Anthropogenic disturbance may lead to the spread of vector-borne diseases through effects on pathogens, vectors, and hosts. Identifying the type and extent of vector response to habitat change will enable better and more accurate management strategies for anthropogenic disease spread. We compiled and analyzed data from published empirical studies to test for patterns among flea and small mammal diversity, abundance, several measures of flea infestation, and host specificity in 70 small mammal communities of five biomes and three levels of human disturbance: remote/wild areas, agricultural areas, and urban areas. Ten of 12 mammal and flea characteristics showed a significant effect of disturbance category (six), biome (four), or both (two). Six variables had a significant interaction effect. For mammal-flea communities in forest habitats (39 of the 70 communities), disturbance affected all 12 characteristics. Overall, flea and mammal richness were higher in remote versus urban sites. Most measures of flea infestation, including percent of infested mammals and fleas/mammal and fleas/mammal species increased with increasing disturbance or peaked at intermediate levels of disturbance. In addition, host use increased, and the number of specialist fleas decreased, as human disturbance increased. Of the three most common biomes (forest, grassland/savanna, desert), deserts were most sensitive to disturbance. Finally, sites of intermediate disturbance were most diverse and exhibited characteristics associated with increased disease spread. Anthropogenic disturbance was associated with conditions conducive to increased transmission of flea-borne diseases.

Michael W Dryden; Patricia A Payne; Smith Vicki; Breanne Riggs; Jacob Davenport; Denise Kobuszewski. 2011. Efficacy of dinotefuran-pyriproxyfen, dinotefuran-pyriproxyfen-permethrin and fipronil-(S)-methoprene topical spot-on formulations to control flea populations in naturally infested pets and private residences in Tampa, FL. *Vet. Parasitol.* 182, 281.

James W Carpenter; Michael W Dryden; Butch Kukanich. 2012. Pharmacokinetics, efficacy, and adverse effects of selamectin following topical administration in flea-infested rabbits. *Am. J. Vet. Res.* 73, 562.

Hanzlicek AS, Harkin KR, Dryden MW, Chun R, Payne PA, Nietfeld JC, Debey BM. Canine Schistosomiasis in Kansas: 5 cases (2000-2009). J. Am. Anim. Hosp. Assoc. 47(6):e95–e102, 2011.

Müller GC, Dryden MW, Revay EE, Kravchenko VD, Broce AC, Hampton K, Junnila A, Schlein Y. Understanding attraction stimuli of *Ctenocephalides felis* for non-chemical control methods. Med. Vet. Entomol. 25(4):413-420.

Dryden M, Carithers D, McBride A, Riggs B, Smith L, Davenport J, Smith V, Payne P, Gross S. A comparison of flea control measurement methods for tracking flea populations in highly infested private residences in Tampa FL, following topical treatment of pets with FRONTLINE® Plus (fipronil/(S)-methoprene). Intern. J. Appl. Res. Vet. Med. 9(4):356-567, 2011.

Ross DH, Arther RG, von Simson C, Doyle V and Dryden MW Evaluation of the efficacy of topically administered imidacloprid + pyriproxyfen and orally administered spinosad against cat fleas (*Ctenocephalides felis*): Impact of treated dogs on flea life stages in a simulated home environment. Parasites & Vectors 2012, 5:192 (7 September 2012).

<http://www.parasitesandvectors.com/content/5/1/192/abstract>

Dryden, M.W., et al., Assessment of owner-administered monthly treatments with oral spinosad or topical spot-on fipronil/(S)-methoprene in controlling fleas and associated pruritus in dogs. Vet. Parasitol. (2012), <http://dx.doi.org/10.1016/j.vetpar.2012.09.003>

Linardi, P. M. 2012. Siphonaptera. In: J. A. Rafael; G. A. R. Melo; C. J. B. de Carvalho; S. A. Casari; R. Constantino. (Org.). Insetos do Brasil. Diversidade e Taxonomia. 1a. ed. Ribeirão Preto/SP: Holos Editora, 2012, p. 689-700.

Linardi, P.M. and Krasnov, B. R. Patterns of diversity and abundance of fleas and mites in the Neotropics: host-related, parasite-related and environment-related factors. Medical and Veterinary Entomology, DOI: 10.1111/j.1365-2915.2012.01025.x

The effects of host-related, parasite-related and environmental factors on the diversity and abundance of two ectoparasite taxa, fleas (Insecta: Siphonaptera) and mites (Acari: Mesostigmata), parasitic on small mammals (rodents and marsupials), were studied in different localities across Brazil. A stronger effect of host-related factors on flea than on mite assemblages, and a stronger effect of environmental factors on mite than on flea assemblages were predicted. In addition, the effects of parasite-related factors on flea and mite diversity and abundance were predicted to manifest mainly at the scale of infracommunities, whereas the effects of host-related and environmental factors were predicted to manifest mainly at the scale of component and compound communities. This study found that, in general, diversity and abundance of flea and mite assemblages at two lower hierarchical levels (infracommunities and component communities) were affected by host-related, parasite-related and environmental factors, and compound communities were affected mainly by host-related and environmental factors. The effects of factors differed between fleas and mites: in fleas, community structure and abundance depended on host diversity to a greater extent than in mites. In addition, the effects of factors differed among parasite assemblages harboured by different host species.

Linardi, P. M. and Santos, J. L. C. *Ctenocephalides felis felis* vs. *Ctenocephalides canis* (Siphonaptera: Pulicidae): some issues in correctly identifying these species. *Revista Brasileira de Parasitologia Veterinária*, 21 (4): 345-354, 2012.

Matsuura, Yu; Kikuchi, Yoshitomo; Xian Ying Meng; Koga, Ryuichi; Fukatsu, Takema. 2012. Novel Clade of Alphaproteobacterial Endosymbionts Associated with Stinkbugs and Other Arthropods. *Applied & Environmental Microbiology*. Jun2012, Vol. 78 Issue 12, p4149-4156. 8p.

Here we report a novel clade of secondary endosymbionts associated with insects and other arthropods. Seed bugs of the genus *Nysius* (Hemiptera: Lygaeidae) harbor the primary gammaproteobacterial symbiont *Schneideria nysicola* within a pair of bacteriomes in the abdomen. Our survey of *Nysius* species for their facultative bacterial associates consistently yielded a novel type of alphaproteobacterial 16S rRNA gene sequence in addition to those of *Wolbachia*. Diagnostic PCR survey of 343 individuals representing 24 populations of four *Nysius* species revealed overall detection rates of the alphaproteobacteria at 77.6% in *Nysius plebeius*, 87.7% in *Nysius* sp. 1, 81.0% in *Nysius* sp. 2, and 100% in *Nysius expressus*. Further survey of diverse stinkbugs representing 24 families, 191 species, and 582 individuals detected the alphaproteobacteria from an additional 12 species representing six families. Molecular phylogenetic analysis showed that the alphaproteobacteria from the stinkbugs form a distinct and coherent monophyletic group in the order Rickettsiales together with several uncharacterized endosymbionts from fleas and ticks. The alphaproteobacterial symbiont clade was allied to bacterial clades such as the endosymbionts of acanthamoebae, the endosymbionts of cnidarians, and *Midichloria* spp., the mitochondrion-associated endosymbionts of ticks. In situ hybridization and electron microscopy identified small filamentous bacterial cells in various tissues of *N. plebeius*, including the bacteriome and ovary. The concentrated localization of the symbiont cells at the anterior pole of oocytes indicated its vertical transmission route through host insect generations. The designation "Candidatus Lariskella arthropodarum" is proposed for the endosymbiont clade.

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Sanchez J., V. Amor, E. A. Bazán-León, R.A. Vásquez & M. Lareschi. 2012. Redescription of *Neotyphloceras chilensis* Jordan, new status (Siphonaptera: Ctenophthalmidae: Neotyphloceratini). *Zootaxa* 3259: 51-57.

Ezquiaga M.C. & M. Lareschi. 2012. Ultrastructure of the Eggs of *Malacopsylla grossiventris* and *Phthiropsylla agenoris* (Siphonaptera, Malacopsyllidae). *Journal of Parasitology* 98 (5): 1029-1031.

The fleas of the Malacopsyllidae are known only by their adults. In this study, we describe the eggs of *Malacopsylla grossiventris* and *Phthiropsylla agenoris* and compare their surface ultrastructure using scanning electron microscopy. Eggs of both flea species are similar in their general appearance,

i.e., the presence of disks and absence of lateral aeropyles, but they differ in size, texture of the surface, and number of micropyles and aeropyles. In addition, the eggs of *M. grossiventris* and *P. agenoris* are larger (719–800 µm) than most of the flea eggs known (<600 µm), and the length/width ratio is 2.0, while it varies between 1.6 and 1.8 in other species. As opposed to other large species, malacopsyllids have expandable (telescoping) abdomens that display growth. It is possible that a clutch of eggs has more than 2 eggs. Females of *M. grossiventris* were observed fixed with their mouthparts to the skin of their hosts. Indeed, some of these specimens were observed copulating on the venter of their hosts. These results contribute to the knowledge about the biology of malacopsyllids.

Nava S. & M. Lareschi. 2012. Ecological characterization of a community of arthropods parasitic of sigmodontine rodents in the Argentinean Chaco. *Journal of Medical Entomology* 49 (6): 1276-1282.

Gozzi A.C., V.V. Benitez, M. L. Guichón & M. Lareschi. 2012. First report of ectoparasites associated with Red Bellied Squirrel *Callosciurus erythraeus* (Rodentia: Sciuridae) introduced into Argentina. *Medical and Veterinary Entomology* (in press).

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Bazán-León E., M. Lareschi, J. Sanchez, G. Soto-Nilo, I. Lazzoni, C. Venegas and R. A. Vásquez. Assemblage of fleas (Insecta: Siphonaptera) associated with non-flying small mammals communities from northern and central Chile; with new host and locality records. *Medical and Veterinary Entomology* (in press).

PILOSOFF, SHAI, MARCELA LARESCHI, and BORIS R. KRASNOV. 2012. Host body microcosm and ectoparasite infracommunities: arthropod ectoparasites are not spatially segregated. *Parasitology* 139, no. 13: 1739-1748.

We studied the distribution of ectoparasite species (an ixodid tick, a chigger mite, 7 mesostigmatid mites, 5 fleas and 3 lice) on bodies of 5 species of rodent hosts from the marshlands in Argentina to establish whether arthropod ectoparasites are segregated across body parts of the same host individual. We asked (a) whether an individual ectoparasite species prefers certain parts of the body of its host and, if yes, whether these preferences overlap among ectoparasite species; (b) whether ectoparasite species composition differs among different parts of a host's body; and (c) whether co-occurrences of ectoparasite species within pre-defined body parts of a host are non-random and, if yes, whether ectoparasites co-occur in the same body part of a host either less or more often than expected by chance. It was found that, in general, ectoparasite species were not segregated across body parts of a host. Although some ectoparasites preferred certain body parts, these preferences were similar among ectoparasites belonging to different species and/or higher taxa resulting in similarity among host body parts in ectoparasite species composition. In addition, ectoparasite species demonstrated a tendency to co-occur on the same body parts of a host and not to be segregated among them. It was concluded that the distribution of ectoparasites on the body of a small mammalian host is driven mainly by their interaction with the host rather than by distinct preferences or interactions among ectoparasite species.

Gislaine Cristina Roma; Patrícia Rosa De Oliveira; Andrea Mendez Araujo; Gervásio Henrique

Bechara; Maria Izabel Camargo Mathias. 2012. Genotoxic and mutagenic effects of permethrin in mice: Micronuclei analysis in peripheral blood erythrocytes. *Microsc. Res. Tech.* 75, 1732.

**

Fleas in Art and History

"Mark but this flea, and mark in this,
 How little that which thou deniest me is ;
 It suck'd me first, and now sucks thee,
 And in this flea our two bloods mingled be.
 Thou know'st that this cannot be said
 A sin, nor shame, nor loss of maidenhead ;
 Yet this enjoys before it woo,
 And pamper'd swells with one blood made of two ;
 And this, alas ! is more than we would do.
 O stay, three lives in one flea spare,
 Where we almost, yea, more than married are.
 This flea is you and I, and this
 Our marriage bed, and marriage temple is.
 Though parents grudge, and you, we're met,
 And cloister'd in these living walls of jet.
 Though use make you apt to kill me,
 Let not to that self-murder added be,
 And sacrilege, three sins in killing three.
 Cruel and sudden, hast thou since
 Purpled thy nail in blood of innocence?
 Wherein could this flea guilty be,
 Except in that drop which it suck'd from thee?
 Yet thou triumph'st, and say'st that thou
 Find'st not thyself nor me the weaker now.
 'Tis true ; then learn how false fears be ;
 Just so much honour, when thou yield'st to me,
 Will waste, as this flea's death took life from thee."

—John Donne, *The Flea* (ca. 1610) in *Poems of John Donne*, vol. 1, pp. 1-2 (E. K. Chambers ed. 1896)

**

"But it is apparent from its applications that 'uku is a root word having the general meaning of ectoparasitic "bug", or an arthropod resembling one.... Fleas ('uku-lele) are thus the leaping ectoparasites (lele = leap, jump, hop, bounce).... The word 'uku-lele, after it was fixed for reference to

household fleas, was adopted as the name of the small 4-string guitar that became a popular musical instrument soon after arrival of Portuguese contract laborers in 1878. The usually accepted version of this adoption, as stated by Webster's dictionaries, is that the rapid movements of the fingers in playing the instrument resemble the leaping of fleas. However, Pukui and Elbert suggest the interesting alternative probability that the ukulele was named for Edward "Uku-lele" Purvis, a small, nimble entertainer who was greatly responsible for popularizing this instrument in Hawaii."

- Haas, G. E., P. Q. Tomich & N. Wilson. (1971). The Flea in Early Hawaii. *Hawaiian Journal of History* 5: 59-74.



The Flea (1707-09, oil on copper, 28x24 cm) by Giuseppe Maria Crespi (1665-1747, Bologna, Italy)

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Siphonaptera Literature (July to December 2012)

Note: please check all citations for accuracy. Some citations list authors in incorrect order, or were published before, or after, the period listed above, or may be repeated in other volumes of Flea News.

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Yoong P, Cywes-Bentley C, Pier GB. 2012. Poly-N-acetylglucosamine expression by wild-type *Yersinia pestis* is maximal at mammalian, not flea, temperatures. *MBio.* Aug 14;3(4):e00217-12.

Zimba M, Loveridge J, Pfukenyi DM, Mukaratirwa S. 2012. Seasonal abundance and epidemiological indices of potential plague vectors *Dinopsyllus lypus* (Siphonaptera: Hystrichopsyllidae) and *Ctenophthalmus calceatus* (Siphonaptera: Ctenophthalmidae) on rodents captured from three habitat types of Hatcliffe and Dzivarasekwa suburbs of Harare, Zimbabwe. *J Med Entomol.* Nov;49(6):1453-9.

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Obituary

Glenn Edward Haas
25-January-1929 • 19-September-2012

Glenn was born in Chicago, Cook County, IL. He received the B.S. from Purdue University, West Lafayette, IN, in 1951, the M.S. from the University of Wisconsin at Madison in 1952 and the Ph.D. in 1957. Both of the latter degrees were in Entomology and his doctoral dissertation was entitled "Ectoparasites of the Mearns cottontail in Wisconsin."

From 1957 to 1959 Haas served as Entomologist to the 9th Naval District, Great Lakes, IL. In 1960 and 1961 he served briefly with Dr. Irving Fox at the University of Puerto Rico before being appointed Animal Ecologist - Medical Entomologist with the Plague Research Unit in the State of HI.

In 1966 and 1967 he was Visiting Assistant Professor in the Department of Zoology in the laboratory of Dr. C.E. Hopla at the University of Oklahoma at Norman, and later appointed to the position of Health Scientist Supervisor and Environmental Scientist with the New Mexico Health and Social Services Department at Los Alamos, NM. From 1971 to 1973 he was Visiting Scientist and Research Associate in the Biology Department at the University of Northern Iowa at Cedar Falls, IA. In August of 1973 he accepted an appointment as Sanitarian in the Environmental Health Section of the Alaska Department of Health and Human Services at Palmer, AK and later at Anchorage, a position he occupied until his retirement in 1980.

He authored or coauthored a number of studies on the fleas of Alaska, and in association with C.E. Hopla (*Flea News* 63 December 2008) and G.P. Holland made major contributions to our knowledge of the Alaskan flea fauna.

Glenn's enthusiasm for entomology continued into recent years. He was a voracious reader and followed college athletics, and his cutting political observations and wry sense of humor were often shared around the campfire. He passed away in Boulder City, NV after an extended period of poor health.

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Photo caption: Glenn Haas in Southern Utah, USA September 2003 (37°31'19"N 114°01' 25"W).

Glenn Haas Bibliography with Selected Abstracts

- Haas, G. E. & R. J. Dicke. (1958). On *Cuterebra horripilum* Clark (Diptera: Cuterebridae) parasitizing cottontail rabbits in Wisconsin. *Journal of Parasitology* 44: 527-540.
- Haas, G. E. & R. J. Dicke. (1959). Fleas collected from cottontail rabbits in Wisconsin. *Transactions of the Wisconsin Academy of Sciences, Arts and Letters* 48: 125-133.
- Haas, G. E. (1965). Comparative suitability of the four murine rodents of Hawaii as hosts for

Xenopsylla vexabilis and *X. cheopis* (Siphonaptera). *Journal of Medical Entomology* 2(1): 75-83.

Host suitability was compared through 10 combinations of rodents with known populations of fleas in simulated nests with the following measurements made: percent of unfed fleas, percent of fleas missing, ratio of fed fleas on host to total fed, and number of adult ♀♀ produced per fed ♀. Adult ♀ rodents ranked in decreasing order of suitability for *XENOPSYLLA VEXABILIS* Jordan as follows: *Rattus exulans* (Peale), *R. norvegicus* (Berkenhout), *R. rattus* (L.), and *Mus musculus* L. In further tests with *R. exulans*, the order was adult ♀, adult ♀, and juvenile ♀. Adult ♀ rodents ranked in decreasing order of suitability for *X. cleopis* (Rothschild) as follows: *R. norvegicus*, *R. rattus*, *R. exulans*, and *M. musculus*.

Haas, G. E. (1965). Another specimen of *Opisocrostitis bruneri* (Baker) with two spermathecae (Siphonaptera). *Journal of Medical Entomology* 2: 140.

Haas, G. E. (1965). Temperature and humidity in the microhabitat of rodent fleas in Hawaiian cane fields. *Journal of Medical Entomology* 2(3): 313-316.

Hawaiian sugar cane fields, in the enzootic plague region, offer a nearly stable temperature and humidity for fleas associated with the infection. In occupied rodent nests in boxes, host body heat raised the temperature above that of the surrounding soil which is determined by climate, not weather, and cane cover. Moderate temperature and high per cent relative humidity in nests were always suitable for breeding by the most abundant rodent flea in the fields, *Xenopsylla vexabilis*.

Haas, G. E. (1966). Cat flea-mongoose relationships in Hawaii. *Journal of Medical Entomology* 2(4): 321-326.

Ecological, behavioral, and physiological evidence is presented to show that the mongoose, *Herpestes auropunctatus auropunctatus* (Hodgson), is a true host of the cat flea, *Ctenocephalides felis* (Bouché), in Hawaii. Monthly samples taken during a 3-year field study in the Hamakua District, Island of Hawaii, showed that the incidence of cat fleas was higher on male than on female mongooses, especially during the mongoose breeding season. In laboratory experiments the sexes were equally suitable as hosts. Thus, the higher flea incidence on males is attributed to their larger body size and to their more intense activity during the breeding season. Very young mongooses were more heavily infested than older ones. Comparative samples of mongooses from the Hamakua, N Kohala, and S Kona Districts showed extraordinarily high host- and flea-population densities in Kona. In data from the entire study, flea sex ratios were not significantly different on male and female hosts. The ratios were also stable suggesting that the fleas bred continuously without significant fluctuations. The mongoose, with its apparently simple denning habits, and the cat flea are sufficiently adapted to each other for a successful association.

Haas, G. E. (1966). A technique for estimating the total number of rodent fleas in cane fields in Hawaii. *Journal of Medical Entomology* 2(4): 392-394.

A technique with subterranean nest boxes and cage traps in combination was developed for estimating the monthly total number of rodent fleas per hectare of a sugar cane field throughout the 22-month crop without excavating burrows for nests. Sampling with replacement of fleas and rodents in nests will provide the mean numbers of fleas and rodents per nest. The estimated number of rodents per hectare

of cane divided by the mean number per nest will give the total number of nests per ha. Multiplication of this number by the mean number of fleas in nests will give the total number of fleas in nests per ha. Multiplication of the number of rodents per ha by the mean number of fleas on trapped rodents will give the total number of fleas on trapped rodents per ha. Then the absolute flea index for each rodent can be obtained by adding fleas in nests and on rodents and dividing by the number of rodents per ha. This technique is subject to errors of uncertain magnitude. Nevertheless, nest boxes have value in studies of flea-nest-host relations.

Haas, G. E. & N. Wilson. (1967). *Pulex simulans* and *P. irritans* on dogs in Hawaii (Siphonaptera: Pulicidae). *Journal of Medical Entomology* 4(1): 25-30.

Pulex irritans L. was believed to be the only species of *Pulex* in Hawaii until *P. simulans* Baker was discovered in 1963 on a dog at Punaluu, Island of Hawaii. Evidence indicates *P. simulans* has been in Hawaii more than 43 years but was confused with *P. irritans*. In 1964, a survey of *Pulex* infesting dogs on Hawaii showed *P. simulans* widespread, but *P. IRRITANS* present only on the northern half of the island where it occurred most frequently at higher elevation, especially at Kukuihaele. The wider distribution of *P. simulans* included the driest, wettest and warmest areas sampled on Hawaii. At least 3 dogs were infested with both species. *P. simulans* was recorded for the first time from Oahu and Kauai, but not Molokai, Lanai or Maui.

Haas, G. E. (1969). Quantitative Relationships between Fleas and Rodents in a Hawaiian Cane Field. *Pacific Science* 23: 70-82.

Relative numbers of *Xenopsylla vexabilis* Jordan on *Rattus exultans* (Peale) and *Mus musculus* L. and in their nests were determined monthly in a field of maturing sugar cane at Kukuihaele, island of Hawaii, January 1962-March 1963, in a preliminary study using nest boxes and live traps. Mean numbers of fleas in active rat nests were most closely correlated with mean numbers of fleas infesting rats when they were cage-trapped in the previous month. Mean numbers of fleas in active mouse nests were most closely correlated with mean numbers of mice that had been recorded per mouse nest found occupied in the previous month. Ratios of mean number of fleas on trapped rats to mean number of fleas in rat nests the following month averaged 0.20 when means increased and 0.30 when they decreased. When the total flea index was substituted for mean number of fleas in nests, ratios averaged 0.32 when fleas became more abundant and 0.36 when they decreased in numbers. Further study of fleas in rat nests is needed before predictions of flea abundance are routinely made from fleas counted on trapped rats.

Haas, G. E. (1970). Rodent fleas in a red fox den in Wisconsin. *Journal of Mammalogy* 51: 796-798.

Haas, G. E., P. Q. Tomich & N. Wilson. (1971). The Flea in Early Hawaii. *Hawaiian Journal of History* 5: 59-74.

Haas, G. E., R. P. Martin, M. Swickard & N. Wilson. (1971). Notes on small mammals infrequently recorded from the Jemez Mountains, New Mexico. *Bulletin of the New Mexico Academy of Science* 12(2): 10-14.

Haas, G. E., R. P. Martin, M. Swickard & N. Wilson. (1972). Bird fleas (Siphonaptera) of New Mexico. *The Canadian Entomologist* 104(6): 881-883.

Only two species of bird fleas were known from New Mexico: *Ceratophyllus celsus celsus* Jordan and *C. petrochelidoni* Wagner, both from the southeastern part of the state. Four new records of the latter species from northcentral New Mexico are added. An unpublished old record and seven new records of *C. niger* Fox are presented. Records of three rodent fleas, *Catallagia decipiens* Rothschild, *Monopsyllus wagneri* (Baker), and *M. eumolpi* (Rothschild), found in bird nests also are given.

Mendez, E. & G. E. Haas. (1972). A new flea of the genus *Megarthroglossus* Jordan and Rothschild from New Mexico (Siphonaptera: Hystrichopsyllidae: Anomiopsyllinae). *Journal of Medical Entomology* 9(4): 285-288.

Megarthroglossus cavernicolus Méndez & Haas, n. sp. of flea from New Mexico, is described and illustrated. This new taxon was found to be sympatric with *M. bisetis* Jordan and Rothschild.

Haas, G. E. (1972). Partial castration in *Monopsyllus vison* (Baker) (Siphonaptera). *Entomological News* 83: 275-278.

Haas, G. E. & P. Q. Tomich. (1972). Ectoparasites of the Hawaiian Islands. I. Siphonaptera. *Contributions of the American Entomological Institute* 8(5): 1-76.

Haas, G. E. & D. W. Walton. (1973). Fleas (Siphonaptera) infesting small mammals from the Western Oriental region. *Korean Journal of Parasitology* 11(2): 102-107.

Haas, G. E. & R. P. Martin. (1973). Bot infestations of pinyon mice in New Mexico. *Entomological News* 84: 89-90.

Haas, G. E. & P. Q. Tomich. (1973). The fleas (Siphonaptera) of Egypt. Early records from mammals. *Journal of the Egyptian Public Health Association* 48: 1-17.

Haas, G. E. (1973). Morphological notes on some Siphonaptera (Leptopsyllidae and Ceratophyllidae) of New Mexico. *American Midland Naturalist* 90: 246-252.

Morphological variations, normal and abnormal, in 11 species of rodent fleas from New Mexico are described and illustrated. *Orchopeas fulleri* Traub is a new United States record, and *Dactylopsylla rara* Fox is a new state record.

Haas, G. E., R. P. Martin, M. Swickard, & B. E. Miller. (1973). Siphonaptera-mammal relationships in northcentral New Mexico. *Journal of Medical Entomology* 10(3): 281-289.

Fifty-two species of fleas were collected from 37 species of mammals, from nests and burrows west of the Rio Grande in and near the Jemez Mts., New Mexico, from November 1969 to August 1971. Data are presented on relative abundance, geographic distribution, seasonal occurrence, host and nest preference, sex ratio, and plague infection. Three species were found naturally infected with plague: *Epitedia stanfordi* Traub for the first time in New Mexico, *Anomiopsyllus princei* Barnes for the first time, and *Monopsyllus wagneri* (Baker). Four species are reported from the state for the first time: *Corrodopsylla c. curvata* (Rothschild), *Myodopsylla gentilis* Jordan & Rothschild, *Dactylopsylla rara* Fox, and *Malaraeus vorifintelis* Prince.

Mendez, E. & G. E. Haas. (1973). *Megarhthroglossus wilsoni*, new species, with notes on the genus in New Mexico (Siphonaptera: Hystrichopsyllidae). *Annals of the Entomological Society of America* 66: 1129-1139.

Haas, G. E., R. E. Barrett & N. Wilson. (1978). Siphonaptera from mammals in Alaska. *Canadian Journal of Zoology* 56(2): 333-338.

Twenty-three species of fleas were collected from 18 species of wild mammals, 2 domesticated species, and man. *Ctenocephalides felis felis* (Bouché) and *Hystrichopsylla dippiei spinata* Holland are new records for the state. *Rhadinopsylla alphabetica* Jameson & Sakaguti is a new record for North America.

Haas, G. E., T. Rumfelt, L. Johnson & N. Wilson. (1979). Siphonaptera from mammals in Alaska. Supplement I. *Canadian Journal of Zoology* 57(9): 1822-1825.

This is the first supplement to Haas *et al.* *Can. J. Zool.* 56: 333–338, 1978. Nineteen species of fleas were collected from 10 species of wild mammals and man. *Delotelis hollandi* Smit and *Ceratophyllus niger* C. Fox. first collected in 1950 and 1937, respectively, were rediscovered. The ranges of four species were extended 500–975 km.

Haas, G. E. & N. Wilson. (1979). Fleas (Siphonaptera) from nests of the cliff swallow (*Petrichelidon pyrrhonota*) in Alaska. *Wasmann Journal of Biology* 37: 59-63.

Haas, G. E., T. Rumfelt, R. E. Barrett & N. Wilson. (1980). Fleas from some Alaskan birds (Siphonaptera). *Pan-Pacific Entomologist* 56: 105-106.

Haas, G. E., T. Rumfelt & N. Wilson. (1980). Fleas (Siphonaptera) from nests and burrows of the Bank Swallow (*Riparia riparia*) in Alaska. *Northwest Science* 54: 210-215.

Wilson, N. & G. E. Haas. (1980). Ectoparasites (Mallophaga, Diptera, Acari) from Alaskan birds. *Proceedings of the Entomological Society of Washington* 82: 541-552.

Haas, G. E. (1980). Unusual Alaskan fleas. P. 347 in R. Traub & H. Starcke (eds.), *Fleas. Proceedings of the International Conference on Fleas*, Ashton Wold, Peterborough, UK, 21-25 June 1977. A. A. Balkema, Rotterdam.

Haas, G. E., L. Johnson & N. Wilson. (1980). Siphonaptera from mammals in Alaska. Supplement II. Southeastern Alaska. *Journal of the Entomological Society of British Columbia* 77: 43-46.

Haas, G. E. (1981). Ectoparasites. Ticks (Acari: Ixodidae). Chewing lice (Mallophaga). Sucking Lice (Anoplura). Fleas (Siphonaptera): Pulicidae and Vermipsyllidae). Pp. 179, 184-186 in R. Barrett and J. Dau (Eds, chapter 3), in R. A. Dieterich (ed.), *Alaskan Wildlife Diseases*. Institute of Arctic Biology, University of Alaska, Fairbanks.

Haas, G. E. & L. Johnson. (1981). A Siphonaptera collection from small mammals by the Canol Road, Yukon Territory. *The Canadian Entomologist* 113(6): 567-567.

Stephen S. Beare, Department of Zoology, University of Alberta, studied martens (*Martes americana*) in the Evelyn Creek area, Nisutlin River drainage, near the Canol Road, 60°47'; 133°15' in 1980 and collected a marten flea and a woodchuck flea from them. In addition, he trapped other small mammals from 15 July to 28 September and collected 109 flea specimens of nine taxa from them.

Haas, G. E., T. Rumfelt & N. Wilson. (1981). Fleas (Siphonaptera) from nests of the tree swallow (*Iridoprocne bicolor*) and the violet-green swallow (*Tachycineta thalassina*) in Alaska. *Wasmann Journal of Biology* 39: 37-41.

Haas, G. E., N. Wilson, R. L. Zarnke, R. E. Barrett & T. Rumfelt. (1982). Siphonaptera from mammals in Alaska. Supplement III. Western Alaska. *Canadian Journal of Zoology* 60(4): 729-732.

Haas, G. E. (1982). Fleas (Siphonaptera) from vole nests in subarctic Alaska. *Canadian Journal of Zoology* 60(9): 2157-2161.

Vole fleas totalling 2420 specimens were obtained from 160 of 291 nests, principally of *Microtus* spp., collected in 66 localities in five regions of Alaska below the Arctic Circle, from 1973 to 1980. Six taxa were in family Hystrichopsyllidae, three in Leptopsyllidae, and five in Ceratophyllidae. Seven taxa are known as Nearctic and six as Amphiberingian, and together in Alaska they form a complex of distribution patterns that was examined from a pentaregional viewpoint. Nearctic forms were preponderant, in numbers of individuals if not taxa, in the southeast, south central, and southwest regions; Amphiberingian forms were preponderant in the interior and especially the west central region. Rearing records suggested that three of the taxa have more than one generation per year. A few shrew, ground squirrel, and ermine fleas were also found in vole nests, and two taxa of vole fleas were reared from lemming nests.

Haas, G. E. (1982). Siphonaptera: the William M. Wallace collection of fleas from Oregon and Washington. *Wasmann Journal of Biology* 40: 45-47.

Haas, G. E. & N. Wilson. (1982). Fleas (Siphonaptera) from nests of the red squirrel (*Tamiasciurus hudsonicus*) and burrows of the arctic ground squirrel (*Spermophilus parryii*) in Alaska. *Wasmann Journal of Biology* 40: 59-65.

Haas, G. E., Wilson, N., Zarnke, R. L., Barrett, R. E., & Rumfelt, T. (1982). Siphonaptera from mammals in Alaska. Supplement III. Western Alaska. *Canadian Journal of Zoology*, 60(4), 729-732.

Nineteen taxa of mammal fleas were recorded from 93 collections from western Alaska. *Oropsylla alaskensis* (Baker) and *O. arctomys* (Baker) are additions to this series of reports. The other 17 taxa were recorded from 71 new localities; 46 records were from remote localities. Many range extensions resulted and noteworthy were *Ctenocephalides f. felis* (Bouché) north to Fairbanks, *Corrodopsylla c. curvata* (Rothschild) west to the Bering Sea coast, *Megabothris abantis* (Rothschild) northwest to the lower Yukon River Valley, and *Chaetopsylla tuberculaticeps* (Bezzi) north of the Arctic Circle. Six hosts were not in our previous records: masked shrew, woodchuck, meadow vole, collared lemming, black bear, and wolverine. In all, 14 species of wild mammals, domestic dog and cat, and man were recorded hosts. Records from bat roosts were included.

Haas, G. E., L. Johnson & R. E. Wood. (1982). Siphonaptera from mammals in Alaska. Supplement IV. Revised check list for southeastern Alaska. *Journal of the Entomological Society of British Columbia* 79: 54-61.

Haas, G. E. (1983). Intersexes and abnormal genitalia in Alaskan Fleas (Siphonaptera). *Journal of the Arizona-Nevada Academy of Science* 18: 4-12.

Intersex specimens of *Catallagia charlottensis* (Baker), *Amphipsylla sibirica pollionis* (Rothschild), *A. marikovskii ewingi* I. Fox, and *Ceratophyllus styx riparius* Jordan & Rothschild are recorded and illustrated. Malformed genitalia in males of *Delotelis hollandi* Smit, *Rhadinopsylla alphabetica* Jameson & Sakaguti, *Peromyscopsylla ostsibirica* (Scalon), *C. s. riparius*, *Ceratophyllus idius* Jordan & Rothschild, and *Megabothris calcarifer gregsoni* Holland and in females of *C. s. riparius* are recorded and illustrated. Parasitic nematodes (order Tylenchida, family Allantonematidae) were confirmed in two of the intersexes and three other abnormal specimens.

Haas, G. E. (1983). Anomalies of the head, thorax and abdomen in the order Siphonaptera. *Advances in Bioscience* 2: 133-144.

Haas, G. E., A. J. Beck & P. Q. Tomich. (1983). Bat fleas (Siphonaptera: Ischnopsyllidae) of California. *Bulletin of the Southern California Academy of Sciences* 82: 103-114.

Tomich, P. Q., A. M. Barnes, W. S. Devick, H. H. Higa & G. E. Haas. (1984). Evidence for the extinction of plague in Hawaii. *American Journal of Epidemiology* 119: 261-273.

Plague (infection by *Yersinia pestis*) was present in Hawaii for the period 1899-1957 and caused at least 370 fatalities. The first infections came from immigrant commensal rats, probably *Rattus rattus* and *Rattus norvegicus*, on ships from the Orient. Both species were already established in Hawaii and became the widespread local carriers of plague, supplemented by *Rattus exulans* which had colonized the islands in ancient Polynesian times. The flea *Xenopsylla vexabilis* arrived with *R. exulans*, and its near relative *Xenopsylla cheopis* accompanied the ship rats. Following each introduction to port cities, plague subsided after a few years but remained active in rural areas of two islands for nearly 50 years. In Hamakua District on Hawaii, the demise of plague was characterized by its repeated terminal migrations from persistent foci, a rapid decline in expected numbers of infected rats and fleas, and negative serologic evidence from rodent reservoirs and mongooses in massive surveys. The simple biotic system - bacterium, three rodents, and two fleas appeared unable to maintain the disease over time. Improved sanitation, mechanized agriculture, gradients in rainfall and temperature, and the collapse of reservoir and vector populations during drought are cited as probable factors in the extinction of plague.

Haas, G. E. & N. Wilson. (1984). Fleas (Siphonaptera) from nests of woodpeckers in Alaska. *Journal of the New York Entomological Society* 92: 125-130.

A total of 955 specimens came from 39 nests in 34 localities west of the Yukon Territory, 1974-1978. Two Holarctic species were *Ornithophaga anomala* Mikulin with 371 specimens from 21 nests (mostly Downy Woodpecker and Three-toed Woodpecker) in 20 localities and *Ceratophyllus zhovtyi* Emel'yanova and Goncharov with 28 specimens from three nests (two Flicker and an unknown species)

in three localities. Two Nearctic species were *C. adustus* Jordan with 168 specimens from 15 nests (mostly Three-toed Woodpecker) in 13 localities and *C. rauschi* Holland with 388 specimens from eight nests (seven Flicker) in six localities.

Haas, G. E. & N. Wilson. (1984). Bird fleas (Siphonaptera: Ceratophyllidae) of the genera *Dasypsyllus* and *Mioctenopsylla* in Alaska. *Wasmann Journal of Biology* 42: 72-77.

Haas, G. E., N. Wilson & T. Rumlft. (1985). Bird fleas, genus *Ceratophyllus*, from Alaska (Siphonaptera: Ceratophyllidae). *Pan-Pacific Entomologist* 61: 180-183.

Wilson, N. & G. E. Haas. (1985). Occurrence of *Hesperocimex coloradensis* List (Hemiptera: Cimicidae) in a woodrat nest in Arizona. *Proceedings of the Entomological Society of Washington* 87: 682.

Haas, G. E., N. Wilson & H. D. Brighton. (1986). Observations on Norway rats, *Rattus norvegicus*, in Kodiak, Alaska. *Canadian Field-Naturalist* 100: 383-385.

Haas, G. E. & N. Wilson. (1986). Rodent fleas (Siphonaptera) in tree cavities of woodpeckers in Alaska. *Canadian field-naturalist* 100(4): 554-556.

Eighteen specimens of rodent fleas were collected from nine nests (40 searched) of woodpeckers in tree cavities in dead stubs. Tree squirrel fleas predominated with 1/2 of the specimens being *Ceratophyllus vision* and 1/3 *Orchopeas caedens durus*. The tree squirrel flea *Tarsopsylla octodecimdentata coloradensis* and vole fleas *Catallagia charlottensis* and *Megabothris abantis* were represented by one specimen each. Studies of total populations of tree squirrel fleas should include tree cavities, even those not containing squirrel nests. Most of the rodent fleas collected from the cavities probably were lost there by visiting Red Squirrels, *Tamiasciurus hudsonicus*, and Martens, *Martes americana*.

Haas, G. E., N. Wilson, T. O. Osborne, R. L. Zarnke, L. Johnson & J. O. Wolff. (1989). Mammal fleas (Siphonaptera) of Alaska and Yukon territory. *Canadian Journal of Zoology* 67(2): 394-405.

Twenty-five taxa (species and subspecies) of mammal fleas were recorded from 127 collections in Alaska. Twenty-one of the taxa were collected in 58 new localities. *Ctenophthalmus pseudagyrtes pseudagyrtes* Baker is new for Alaska, and *Microtus xanthognathus* is a new host for this flea. *Chaetopsylla tuberculiceps* (Bezzi) was collected north of the Brooks Range for the first time, and in southeastern Alaska it was found breeding in a bear den. Five records of fleas from nests of shrews, voles, and tree squirrels are included, but most collections were from 17 species of wild mammals, two commensal rodents, domestic dog and cat, and man. Six records of six taxa of rodent fleas from Yukon Territory were added. Checklists for Alaska and Yukon Territory and distribution maps for 16 taxa are provided.

Galloway, T. D., G. E. Haas & A. H. Benton. (1989). Corrections and additions to G. P. Holland's "Fleas of Canada, Alaska and Greenland (Siphonaptera)". *The Canadian Entomologist* 121(07): 609-618.

George Holland's superb monograph on the fleas of Canada, Alaska and Greenland was published as *Memoir of the Entomological Society of Canada* No. 130 on 11 March 1985 (not in 1984 as indicated

on the spine of the monograph). This work was immediately adopted by those of us working with the northern North American flea fauna. Because of George's lengthy illness, he was unable to correct galley proofs to his usual high standards. Unfortunately, George Holland passed away later in 1985, before he was able to address corrections to the monograph. What follows is a compilation of the errors that we have detected since the monograph was published. We have made no attempt to reinterpret Holland's taxonomic or biogeographical concepts, nor in any way change the meaning of the text. We have included additional references that pertain to the relevant geographic areas, either where Holland did not refer to them in the text, or which appeared subsequent to completion of the manuscript, up to and including year of publication (1985).

Haas, G. E. & N. Wilson. (1998). *Polygenis martinezbaezi* (Siphonaptera: Rhopalopsyllidae) reared from a rodent nest found in the Peloncillo Mountains of southwestern New Mexico. *Journal of Medical Entomology* 35(4): 431-432.

A 3rd species of the flea genus *Polygenis* Jordan was added to the U.S. fauna by extension of known range from Mexico into New Mexico. Eleven males and 24 females of *P. martinezbaezi* Vargas were reared from a rodent nest, probably that of a yellow-nosed cotton rat, *Sigmodon ochrognathus* Bailey.

J. R. Kucera, G. E. Haas & M. K. MacDonald. (2001). Fleas (Siphonaptera) from sciurid and murid rodents on the eastern slope of the Cascade Range, Kittitas County, Washington. *Journal of the Entomological Society of British Columbia* 98: 227-233.

Haas, G. E., N. Wilson & C. T. McAllister. (2004). Fleas (Siphonaptera: Ceratophyllidae, Ctenophthalmidae) from rodents in five Southwestern states. *Western North American Naturalist* 64(4): 514-517.

Nine species of fleas were collected incidental to a survey of rodents for molecular studies in Arizona, Colorado, New Mexico, Texas, and Utah, south of latitude 38° N. Three species were parasites of woodrats, *Neotoma* spp.: *Orchopeas neotomae* Augustson was confirmed for Texas, and the distribution patterns of *O. agilis* (Rothschild) and *O. schisintus* (Jordan) were more clearly defined. Four species were parasites of mice, *Peromyscus* spp.: *Aetheca wagneri* (Baker) was a new flea for *P. gratus* Merriam, the distribution of *O. leucopus* (Baker) was extended to far west Texas, *Plusaetis sibynus* (Jordan) was new for Utah and *N. lepida*, and the range of *Stenoponia americana* (Baker) was extended west of the Continental Divide in New Mexico. Other species included *Foxella ignota* (Baker) and *Meringis dipodomys* Kohls.

Haas, G. E., & J. R. Kucera. (2004). Fleas (Siphonaptera) in nests of voles (*Microtus* spp.) in montane habitats of three regions of Utah. *Western North American Naturalist* 64(3): 346-352.

As a parallel study of a survey of fleas of trapped small mammals in montane southern Utah, we removed 77 adult fleas from 12 of 13 nests of voles (*Microtus longicaudus*, *M. montanus* and *M. richardsoni*) collected from montane meadows in the Abajo and La Sal Mountains, the Markagunt Plateau and Pine Valley Mountains, and the Uinta Mountains, May 1991-July 2002. Six species and subspecies of fleas parasitic on *Microtus* spp. were found, 1-3 species in each nest. Three specimens of the ubiquitous deer mouse (*Peromyscus* spp.) flea *Aetheca wagneri* (Baker) were also found. We collected the following 7 taxa from the locations indicated: *Catallagia decipiens* Rothschild in 5 nests: Abajo and Pine Valley Mountains and Markagunt Plateau; *Hystrichopsylla dippiei truncata* Holland in

2 nests: La Sal and Uinta Mountains; *H. occidentalis sylvaticus* Campos & Stark in 5 nests: Pine Valley Mountains; *Peromyscopsylla selenis* (Rothschild) in 1 nest: Markagunt Plateau; *A. wagneri* in 2 nests: Abajo and Pine Valley Mountains; *Megabothris abantis* (Rothschild) in 5 nests: Abajo and Uinta Mountains and Markagunt Plateau; *Malaraeus telchinus* (Rothschild) in 2 nests: Abajo and Pine Valley Mountains. One preserved larva was identified as *Hystrichopsylla* prob. *occidentalis sylvaticus*, and 2 dead larvae, the only specimens found in nest 13, were identified as probably the same. Kane County is a new record for *C. decipiens* and *P. selenis*; Kane and Summit Counties are new for *M. abantis*. Nest surveys can generally supplement rather than replace trapped-host surveys. Their main value is in population studies. They are also a source of larvae for morphology and taxonomy research.

Haas, G. E., J. R. Kucera, A. M. Runck, S. O. MacDonald & J. A. Cook. (2005). Mammal fleas (Siphonaptera: Ceratophyllidae) new for Alaska and the southeastern mainland collected during seven years of a field survey of small mammals. *Journal of the Entomological Society of British Columbia* 102: 65-76.

Ten taxa of mammal fleas were among 124 collection records from 12 host species (one shrew, nine rodents and two carnivores), at 72 localities on the southeastern Alaska mainland in 1989 and during an extensive survey of mammals in 1992-1995 and 1997-1999. *Megabothris asio megacolpus* (Jordan) ex *Microtus pennsylvanicus* (Ord), *Malaraeus telchinus* (Rothschild) ex *Peromyscus keeni* (Rhoads) and *Clethrionomys gapperi* (Vigors) are new fleas for Alaska. *Orchopeas caedens* (Jordan) ex *Tamiasciurus hudsonicus* (Erxleben) is a new flea for southeastern Alaska. *Synaptomys borealis* (Richardson) is a new host record for *Opisodasys k. keeni* (Baker). The other six taxa of fleas collected were *Hystrichopsylla dippiei spinata* Holland, *H. o. occidentalis* Holland, *Catallagia charlottensis* (Baker), *Ceratophyllus ciliatus protinus* Jordan, *Megabothris abantis* (Rothschild) and *Opisodasys vesperalis* (Jordan). Of these, *H. o. occidentalis*, *C. charlottensis* and *M. abantis* have seven new host records for the southeastern Alaska mainland. Distribution patterns of the fleas and their host relationships in North America are discussed.

Hastriter, M. W. & G. E. Haas. (2005). Bionomics and distribution of species of *Hystrichopsylla* in Arizona and New Mexico, with a description of *Hystrichopsylla dippiei obliqua*, n. ssp. (Siphonaptera: Hystrichopsyllidae). *Journal of Vector Ecology* 30(2): 251.

More than 450 specimens of *Hystrichopsylla* were collected from nests and hosts of species of *Microtus*, *Neotoma*, *Tamiasciurus*, and *Peromyscus* in Arizona and New Mexico from 1981-2004. A new subspecies, *Hystrichopsylla dippiei obliqua*, is described and a map illustrating the distribution of the three taxa (*Hystrichopsylla dippiei truncata* Holland, *H. d. obliqua*, and *H. occidentalis sylvaticus* Campos and Stark) occurring in Arizona and New Mexico is provided. *Hystrichopsylla. o. sylvaticus* is reported in New Mexico for the first time and *H. d. truncata* is a new record in Rio Arriba County, NM. Relationships of Mexican species are also discussed. These large fleas are seldom collected from the fur of their mammalian hosts (usually singly) but are prevalent in moist nests. The maximum number of the new subspecies collected from a single nest was 54. Dry nests, particularly those that are not subterranean or otherwise protected from desiccation, do not support development of *Hystrichopsylla* fleas. Minimum elevation at which *H. dippiei* ssp. is found in Arizona and New Mexico appears to be about 2,100 m. *Journal of Vector Ecology* 30 (2): 251-262. 2005.

Haas, G. E. & J. R. Kucera. (2010). Shrew Flea, *Corrodopsylla curvata*, in Nest and Range Extension in New Mexico with Review of Relevant Records in Six Western States. *Southwestern Entomologist*

35(4): 525-532.

The shrew flea, *Corrodopsylla curvata curvata* (Rothschild, 1915) (Siphonaptera: Ctenophthalmidae), is not common in collections of fleas from the Southwest or Rocky Mountains. Fewer records are from nests than trapped hosts. In 1993, a live female specimen was found in a nest that belonged to an unknown small mammal in the southern Sangre de Cristo Mountains of New Mexico. A 67km extension of range was measured from the nearest of four trap sites in the Jemez Mountains, Sandoval County. The nest site was east-southeast across the Rio Grande in Dalton Canyon, Pecos River watershed, San Miguel County. The nest, its contents, and habitat were described. In addition, we reviewed nonpublished ecological data from each of the three collections from trapped hosts in the Jemez Mountains, 1970–1971, and provided a distribution map. A review of records of *C. c. curvata* and *C. c. obtusata* (Wagner, 1929) in western mountains with emphasis on small samples from a canyon in Montana suggested that larger samples would confirm the absence of a zone of intergradation between two allopatric species.

Haas, G. E., N. Wilson, J. R. Kucera, T. O. Osborne, J. S. Whitman & W. N. Johnson. (2010). Range expansion and hosts of *Ctenophthalmus pseudagyrtes* Baker (Siphonaptera: Ctenophthalmidae) in central Alaska. *Journal of the Entomological Society of British Columbia* 107: 87-88.

Haas, G. E. (2012). Large Flea Larvae of *Hystrichopsylla dippiei* Reveal Breeding in Red Squirrel Nest with Adult *H. occidentalis sylvaticus* Confirming Sympatry in New Mexico. *Southwestern Entomologist* 37(2): 151-157.

Sympatry was noted with *Hystrichopsylla dippiei* Rothschild and *H. occidentalis sylvaticus* Campos & Stark and earlier in New Mexico with *Peromyscopsylla selenis* (Rothschild) and *P. hamifer vigens* (Jordan), 1970s. Most significant was the recent occurrence of *H. dippiei* larvae and *H. o. sylvaticus* adults in the same nest of a red squirrel, *Tamiasciurus hudsonicus* (Erxleben) RN-131 in the San Francisco Mountains, Catron County, Mogollon Volcanic Plateau, 23 October 1995. Collections on 20 August 1993, 20 September and 23 October 1995, 1 October 1998, and 1 and 4 October 1999 recorded nine nests of red squirrels and two nests of Mexican woodrats, *Neotoma mexicana* Baird, the latter with visits by deer mice, *Peromyscus maniculatus* (Wagner) or *P. gratus* Merriam. Red squirrel nests had one to four species of adult fleas including one of a woodrat. Adult species totaled six: *H. o. sylvaticus*, *Catallagia decipiens* Rothschild, *Anomiopsyllus nudatus* (Baker), *Megarhthroglossus* aff. *bisetis* Jordan & Rothschild, *Orchopeas caedens* (Jordan), *Orchopeas neotomae* Augustson. Mexican woodrat nests had six and seven species of adult fleas including mouse and red squirrel species. Adults totaled nine: *H. o. sylvaticus*, *C. decipiens*, *A. nudatus*, *M. aff. bisetis*, *Aetheca wagneri* (Baker), *Malaraeus telchinus* (Rothschild), *O. caedens*, *O. neotomae*, and *Plusaetis sibynus* (Jordan).

Haas, G. E., J. R. Kucera, S. O. MacDonald & J. A. Cook. (2012). First flea (Siphonaptera) records for Kanuti National Wildlife Refuge, Central Alaska. *Journal of the Entomological Society of British Columbia* 109: 6-11.

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